SIEMENS S7-1500 · SCL FUNCTION BLOCK · MULTI-INSTANCE IDB INSIDE PPC_CONTROLLER

FB_PPC_QCapability

Plant-level reactive power generation: P-Q capability envelope (linear interpolation from 5-point table), voltage droop Q mode, and Q ramp rate limiting. Called from FB_PPC_Controller step ⑤. ANRE articles 147, 150, 152, 160, 163 — ISCE Tests 6, 8, 9, 10.

VERSION	IDB ACCESS PATH	CALLED FROM
`0.1`	`"PPC_Controller".QCap_IDB.*`	`FB_PPC_Controller step ⑤`

OUTPUT CONSUMER	ANRE ARTICLES
`FC_PPC_ReactiveControl step ⑦`	`147, 150, 152, 160, 163`

Q SIGN CONVENTION & CONTROL MODES

+Q CAPACITIVE / LEADING

Positive kVAr = **overexcited** (capacitive). Plant injects reactive power into the grid, raising voltage. Clamped by `Q_cap_max` rows in PQ_Table.

Example: Q_setpoint_final = +5 000 kVAr → each inverter receives a positive kVAr setpoint.

−Q INDUCTIVE / LAGGING

Negative kVAr = **underexcited** (inductive). Plant absorbs reactive power from the grid, lowering voltage. Clamped by `Q_ind_max` rows (stored positive; sign applied in clamp).

Example: Q_setpoint_final = −5 000 kVAr → each inverter receives a negative kVAr setpoint.

Q=0 ZERO-P Q (ART. 150)

Q dispatch is **not gated on P>0**. When solar irradiance is zero and WSpt=0, inverters with OperMode=308 can still source/sink reactive power. `Control_Mode=2` sets Q=0; modes 0 and 1 work at any P level including P=0.

ISCE Test 10 verifies this explicitly.

0

Fixed Q Setpoint

`Q_setpoint_ext → ramp → P-Q clamp → Q_setpoint_final`

SCADA writes a kVAr target directly. Ramp rate limits apply. Used for ISCE Test 8 (P-Q table verification) and Test 6 (ramp rates).

1

Voltage Droop

`Q = (dU_pu / (U_Droop_pct/100)) × QmaxPlant`

Closed-loop voltage regulation. Q is computed from the per-unit voltage error every OB30 cycle. Used for ISCE Test 9.

2

Off (Q = 0)

Q_setpoint_final = 0. Used when `Plant_VArMode=2` (power factor mode) in FC_PPC_ReactiveControl, which overrides Q dispatch entirely. The ramp still tracks zero gracefully.

CORE FORMULAS `computed every OB30 scan cycle`

P-Q CAPABILITY TABLE — LINEAR INTERPOLATION (ISCE TEST 8)

```
P_pct = (P_actual_kW / Pmax_kW) × 100.0 [%]
tbl_idx = FLOOR(P_pct / 25), clamped to [0, 3] [row index]
t0 = (P_pct − tbl_idx × 25) / 25 [0..1 fractional position]
Q_ind_interp = Table[tbl_idx].Q_ind_max + t0 × (Table[tbl_idx+1].Q_ind_max − Table[tbl_idx].Q_ind_max)
Q_cap_interp = Table[tbl_idx].Q_cap_max + t0 × (Table[tbl_idx+1].Q_cap_max − Table[tbl_idx].Q_cap_max)
```

Then clamped to physical inverter capacity: `MIN(Q_ind_interp, QmaxPlant)`, `MIN(Q_cap_interp, QmaxPlant)`
Example: P_actual=24 000 kW, Pmax=48 000 kW → P_pct=50% → tbl_idx=2, t0=0 → interpolated value = Table[2] exactly.

VOLTAGE DROOP Q COMMAND — MODE 1 (ISCE TEST 9)

```
dU_pu = (U_setpoint_ext − U_meas) / U_meas [dimensionless per-unit]
Q_command = (dU_pu / (U_Droop_pct / 100.0)) × QmaxPlant [kVAr]
```

Guard: if `U_meas ≤ 0` or `U_Droop_pct ≤ 0` → Q_command = 0 (division protection).
Positive voltage error (U_spt > U_meas, under-voltage) → positive Q_command (capacitive injection, raises voltage).
Example: U_Droop_pct=5%, U_meas=20.0 kV, U_spt=20.5 kV, QmaxPlant=15 000 kVAr:
dU_pu = 0.5/20.0 = 0.025 → Q = (0.025/0.05) × 15 000 = **+7 500 kVAr**

Q RAMP RATE LIMITING (ISCE TEST 6)

```
Q_ramp_rate = Q_Ramp_Rate_fast if Q_Ramp_Fast_Sel = TRUE
Q_Ramp_Rate_slow if Q_Ramp_Fast_Sel = FALSE
maxStep = Q_ramp_rate × CycleTime_s [kVAr per scan]
Ramps_Qcmd += CLAMP(target − Ramps_Qcmd, −maxStep, +maxStep)
```

Symmetric ramp — same rate for increasing and decreasing Q (Q can be negative).
In FALLBACK mode (Plant_Mode=2): target forced to 0.0 → Ramps_Qcmd ramps toward zero, no step-change.
Example: Ramp_Rate=500 kVAr/s, CycleTime=0.1 s → maxStep = 50 kVAr per scan.

VAR_INPUT `13 parameters — passed by FB_PPC_Controller from DB39 / InverterMonitor outputs`

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST VALUES
P_actual_kW	REAL	Plant actual active power output in kW, from the grid meter (not a setpoint). Used to determine the current P tier in the P-Q capability table. Should come from the metering system — if meter is not yet wired, use Sim_Mode + a fixed forced value.	0.0 … 48 000 kW	Test 8 tiers: 0, 12 000, 24 000, 36 000, 48 000
Pmax_kW	REAL	Plant rated active power in kW (= Pn_MW × 1000). Divisor for P_pct computation. Must match DB39.Pn_MW × 1000. Never zero — guarded internally (P_pct = 0 if Pmax_kW ≤ 0).	48 000 kW (CEF Tandarei)	48 000.0
Q_setpoint_ext	REAL	External Q command from SCADA or AGC in kVAr. Active only in Control_Mode = 0 (Fixed Q). Positive = capacitive. This is the raw target; it is still ramped and capability-	−20 000 … +20 000 kVAr (within table limits)	Test 8: step through ±25%, ±50%, ±75%, ±100% of Q_cap / Q_ind limit

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST VALUES
		clamped before reaching inverters.		

U_setpoint_ext	REAL	Voltage setpoint from SCADA or AGC in kV. Active only in Control_Mode = 1 (Voltage droop). Typically the MV nominal voltage (20 kV for CEF Tandarei). SCADA steps this during Test 9 to verify droop response.	`0.0 … 30 kV`
			`Nominal: 20.0 kV`
				`Test 9 steps:`
				`20.0 (nominal)`
				`20.5 (+2.5%)`
				`21.0 (+5%)`

U_meas	REAL	MV busbar measured voltage in kV from the AI input (VT on MV bus or grid meter). Used as denominator in voltage droop: `dU_pu = (U_spt - U_meas) / U_meas`. Must be non-zero (internally guarded).	`0.1 … 30 kV` `Typical: 19.5 … 20.5 kV`	`Test 9: fix at 20.0 then step U_setpoint_ext`

Control_Mode	INT	Selects Q generation source for this FB. **0** = Fixed Q (Q_setpoint_ext). **1** = Voltage droop. **2** = Off (Q=0). Independent of Plant_VArMode in FC_PPC_ReactiveControl — this input governs the Q generation source; ReactiveControl governs per-inverter distribution mode.	`0 = Fixed Q`
			`1 = Voltage droop`
			`2 = Off`
				`Test 8: 0`
				`Test 9: 1`
				`Test 10: 0`

U_Droop_pct	REAL	Voltage droop percentage. Defines how many % of per-unit voltage error produces full Q response. 5% means a ±5% voltage	`1.0 … 20.0 %` `Default: 5.0 %`	`Test 9: 5.0`

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST VALUES
		deviation = ±100% QmaxPlant. ANRE Art. 160 minimum. SCADA-configurable — stored in DB39. Only active when Control_Mode = 1.		
Q_Ramp_Rate_fast	REAL	Fast Q ramp rate in kVAr/s. ISCE Test 6 fast-ramp sub-test. SCADA-writable via DB39. Typical value: step response at ≥1% Pn/s ≈ 480 kVAr/s for 48 MW plant.	`100 … 5 000 kVAr/s` `Typical: 500 kVAr/s`	`Test 6 fast: 500.0`
Q_Ramp_Rate_slow	REAL	Slow Q ramp rate in kVAr/s. ISCE Test 6 slow-ramp sub-test. Must be less than Q_Ramp_Rate_fast. SCADA-writable.	`10 … 1 000 kVAr/s` `Must be < Q_Ramp_Rate_fast`	`Test 6 slow: 100.0`
Q_Ramp_Fast_Sel	BOOL	Selects which ramp rate is active. **FALSE** = slow rate (normal operation). **TRUE** = fast rate. SCADA-switchable during Test 6 without code reload. Same scan cycle, no delay.	`FALSE = slow` `TRUE = fast`	`Test 6 fast: TRUE` `Test 6 slow: FALSE`
QmaxPlant	REAL	Total available reactive capacity in kVAr from FC_PPC_InverterMonitor. Sum of VArAval across all available inverters. Used as the Q ceiling for voltage droop gain and as belt-and-suspenders clamp after P-Q table interpolation. Changes dynamically with available inverters.	`0.0 … 48 000 kVAr`	`Sim_Mode: 10 000 per inverter = 100 000 kVAr total (auto-clamped by table)`

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES	ISCE / TEST VALUES
CycleTime_s	REAL	OB30 scan cycle time in seconds. Used to convert ramp rate (kVAr/s) to per-scan increment: `maxStep = Q_ramp_rate × CycleTime_s`. Must match the OB30 cycle time. Default 0.1 s (100 ms OB30).	`0.05 … 1.0 s` `Default: 0.1 s`	`0.1`

Plant_Mode	INT	Current plant operating mode from FC_PPC_ModeManager. When Plant_Mode = 2 (FALLBACK): Q ramp target is forced to 0.0, so Ramps_Qcmd ramps to zero gracefully rather than jumping. Control_Mode is still used to compute Q_command (the effective target override takes priority).	`0 = LOCAL`
			`1 = REMOTE_IEC`
			`2 = FALLBACK`
				`Normal: 1`
				`FALLBACK test: 2`

VAR_OUTPUT `4 parameters — all SCADA-relevant, mirrored to DB39`

PARAMETER	TYPE	DESCRIPTION	NORMAL / ALARM VALUES	DB39 MIRROR FI
Q_setpoint_final	REAL	Plant-level Q command after ramp and P-Q capability clamp, in kVAr. Fed to FC_PPC_ReactiveControl as `Q_cmd_kVAr`. ReactiveControl only distributes this proportionally — no further Q limit logic runs there. **This is the value actually**	`-Q_ind_interp … +Q_cap_interp = 0 when Control_Mode=2 or FALLBACK`	`Visible at inverter leve as VArSpt. No directly in DB39 — monito at ReactiveCont output.`

PARAMETER	TYPE	DESCRIPTION	NORMAL / ALARM VALUES	DB39 MIRROR FI
		dispatched to inverters.		
`Q_max_inductive`	REAL	Interpolated inductive (lagging/underexcited) Q limit at the current P operating point, in kVAr. Always positive — the actual lower bound for Q_setpoint_final is −Q_max_inductive. Changes every cycle with P_actual_kW. SCADA	`0.0 … 20 000 kVAr Positive value always`	`"PPC_Controlle .Q_max_inducti`
`Q_max_capacitive`	REAL	Interpolated capacitive (leading/overexcited) Q limit at the current P operating point, in kVAr. Always positive — the actual upper bound for Q_setpoint_final is +Q_max_capacitive. SCADA	`0.0 … 20 000 kVAr Positive value always`	`"PPC_Controlle .Q_max_capacit`
`Q_limited`	BOOL	TRUE when Q_setpoint_final was clamped by the P-Q capability envelope (ramped Q exceeded table limit at current P). Diagnostic alarm — indicates the SCADA Q command exceeds what the plant can legally deliver at this P level. SCADA ALARM	`FALSE = no clamp TRUE = capability limited`	`"PPC_Controlle .Q_limited`

IDB STATIC VARIABLES (VAR) `retained between OB30 cycles — "PPC_Controller".QCap_IDB.*`

These variables live in the **PPC_Controller instance DB** under `"PPC_Controller".QCap_IDB`. Access directly in the Watch table or via SCADA. `Ramps_Qcmd` can be forced to zero in the Watch table to reset the Q ramp accumulator without a PLC restart.

VARIABLE	TYPE	DESCRIPTION & SIGNIFICANCE	NORMAL STATE	HOW TO ACCESS
Ramps_Qcmd	REAL IDB	Q ramp accumulator in kVAr. The current ramped Q value before capability clamping. Accumulates at the rate set by Q_Ramp_Rate_fast or _slow. Retains its value between cycles (integrator state). WRITABLE Force to 0.0 in Watch table to instantly reset the Q integrator — useful after a mode switch or test step. In normal operation it tracks Q_setpoint_final closely (difference only during ramp transients).	≈ Q_setpoint_final Difference during ramp Zero in FALLBACK	"PPC_Controller .QCap_IDB .Ramps_Qcmd
PQ_Table[0..4]	ARRAY IDB	5-element array of `UDT_PQ_CapPoint` structs. Each element has three REALs: • `P_pct` — % of Pmax for this tier (configure as 0, 25, 50, 75, 100) • `Q_ind_max` — kVAr inductive limit at this tier (positive value) • `Q_cap_max` — kVAr capacitive limit at this tier (positive value) **Default: 20 000 kVAr at all tiers** — non-limiting until commissioned. Replace with inverter datasheet values before ISCE Test 8. The P_pct field is informational only; the interpolation assumes fixed 25% steps (index × 25%). MUST CONFIGURE BEFORE TEST 8	See commissioning table below	"PPC_Controller .QCap_IDB .PQ_Table[0].Q_ etc.

Configure `PQ_Table[0..4]` in the Watch table or via SCADA/HMI **before** ISCE Test 8. Values below are illustrative ANRE Category D minimums (±33% Pn at all P tiers). Replace with actual inverter datasheet values — contact SMA for the specific SCC curve for your inverter model.

INDEX	P_PCT (%)	Q_IND_MAX (KVAR) LAGGING / UNDEREXCITED	Q_CAP_MAX (KVAR) LEADING / OVEREXCITED	IDB WRITE PATH
PQ_Table[0]	0	15 840	15 840	.QCap_IDB.PQ_
PQ_Table[1]	25	15 840	15 840	.QCap_IDB.PQ_
PQ_Table[2]	50	14 000	14 000	.QCap_IDB.PQ_
PQ_Table[3]	75	11 000	11 000	.QCap_IDB.PQ_
PQ_Table[4]	100	9 000	9 000	.QCap_IDB.PQ_

Important: Values above are illustrative only. Obtain the actual P-Q capability diagram (SCC curve) from SMA for your specific inverter model and operating voltage. Values must match the plant single-line diagram and inverter type test reports submitted to ANRE.

SCADA MONITORING MAP `minimum tag set for ANRE ISCE commissioning campaign`

ISCE TEST 8

P-Q Capability Diagram

`Control_Mode (DB39)` Set 0 (Fixed Q)

ISCE TEST 9

Voltage Droop Q Control

`Control_Mode (DB39)` Set 1 (Voltage droop)

`P_actual_kW`	Step 0 / 12k / 24k / 36k / 48k kW
`Q_max_inductive (DB39)`	Record at each P tier
`Q_max_capacitive (DB39)`	Record at each P tier
`Q_setpoint_ext (DB39)`	Step to ±Q_max at each tier
`Q_limited (DB39)`	Must be FALSE at table limit
`Q_limited (DB39)`	Must be TRUE above table limit

`U_Droop_pct (DB39)`	Set 5.0 %
`U_meas`	Fix at 20.0 kV (or measure)
`U_setpoint_ext (DB39)`	Step 20.0 → 20.5 → 21.0
`Q_setpoint_final`	Verify ≈ 0 → +7500 → +15000
`Q_max_capacitive`	Must not be binding

ISCE TEST 10

Zero-P Reactive Capability

`P_actual_kW`	Must be 0 (night / shutter)
`Control_Mode (DB39)`	Set 0 (Fixed Q)
`Q_setpoint_ext (DB39)`	Step ±Q_max_at_P0
`Q_setpoint_final`	Verify dispatched to inverters
`Q_max_inductive/cap`	Must equal PQ_Table[0] values

Inverter OperMode must be 308 (remote Q) — check FaultHandler config

ISCE TEST 6

Q Ramp Rates

`Q_Ramp_Rate_fast (DB39)`	Set target value (e.g. 500)
`Q_Ramp_Rate_slow (DB39)`	Set target value (e.g. 100)
`Q_Ramp_Fast_Sel (DB39)`	Toggle TRUE/FALSE per sub-test
`QCap_IDB.Ramps_Qcmd`	Watch trend — verify slope
`Q_setpoint_final`	Trend — lags Ramps_Qcmd by clamp

Reset Ramps_Qcmd=0 in Watch between sub-tests for a clean step

CONTINUOUS MONITORING

Normal Operation

`Q_limited (DB39)`	Alarm if TRUE (check AGC command)
`Q_max_inductive (DB39)`	Trend — tracks P changes
`Q_max_capacitive (DB39)`	Trend — tracks P changes
`Q_setpoint_final`	Trend — reactive dispatch
`Control_Mode (DB39)`	Display — 0/1/2

WATCH TABLE — DEBUG

IDB Direct Access

`.QCap_IDB.Ramps_Qcmd`	Force 0.0 to reset ramp accumulator
`.QCap_IDB.PQ_Table[0..4].*`	Write table values before Test 8

Prefix all paths with `"PPC_Controller"`